

Comprehensive Evaluation of Model Fusion Strategies for Clinical Prediction Tasks

1. Hospital Mortality (mort_hosp) Analysis

Source Data: mort_hosp_confidence_intervals_summary.md

The baseline XGBoost model for this task demonstrates exceptionally high performance (AUROC > 0.91), setting a challenging benchmark for any fusion strategy to surpass.

Performance Comparison Table: Hospital Mortality (mort_hosp)

Model / Strategy	AUROC (95% CI)	AUPRC (95% CI)
Baseline: XGBoost	0.9107 (0.8999 - 0.9219)	0.6143 (0.5772 - 0.6500)
Baseline: Embedding Model	0.8389 (0.8248 - 0.8532)	0.3872 (0.3497 - 0.4265)
Best Fusion: Blending	0.9100 (0.8968 - 0.9204)	0.6151 (0.5781 - 0.6516)
Hierarchical Uncertainty	0.9088 (0.8974 - 0.9199)	0.5945 (0.5585 - 0.6286)
Hierarchical Confidence	0.9088 (0.8975 - 0.9190)	0.5945 (0.5565 - 0.6307)
Uncertainty-weighted	0.9085 (0.8972 - 0.9202)	0.5999 (0.5627 - 0.6346)
Stacking	0.9048 (0.8931 - 0.9159)	0.6049 (0.5684 - 0.6401)
Confidence-weighted	0.9015 (0.8893 - 0.9126)	0.5906 (0.5524 - 0.6272)
Difficulty-adaptive	0.8951 (0.8824 - 0.9067)	0.5777 (0.5376 - 0.6158)
Edge-enhanced	0.8940 (0.8810 - 0.9059)	0.5745 (0.5338 - 0.6140)
Consistency-weighted	0.8939 (0.8811 - 0.9055)	0.5761 (0.5359 - 0.6148)
Neural Network	0.8734 (0.8596 - 0.8864)	0.6045 (0.5679 - 0.6402)

Analysis and Conclusion

- Overall Trend:** The majority of fusion strategies failed to outperform the powerful

XGBoost baseline. Several strategies, including Neural Network and Edge-enhanced fusion, resulted in a notable degradation in performance.

- **Best Performance:** The **Blending** strategy achieved a negligible improvement in AUPRC (+0.0008) but a slight decrease in AUROC.
- **Statistical Significance:** No fusion strategy demonstrated a statistically significant improvement. The confidence intervals of the top-performing Blending strategy show extensive overlap with the XGBoost baseline, suggesting the minor advantage is likely due to random chance.
- **Recommendation:** The **XGBoost baseline model** is the optimal choice for this task. The fusion strategies do not provide meaningful performance gains and introduce unnecessary complexity.

2. 30-Day Readmission (readmission_30) Analysis

Source Data: readmission_30_confidence_intervals_summary.md

This is a highly challenging task due to the extremely low positive class ratio (4.69%), resulting in modest absolute performance across all models.

Performance Comparison Table: 30-Day Readmission (readmission_30)

Model / Strategy	AUROC (95% CI)	AUPRC (95% CI)
Baseline: XGBoost	0.5973 (0.5645 - 0.6334)	0.0657 (0.0577 - 0.0782)
Baseline: Embedding Model	0.5800 (0.5444 - 0.6152)	0.0632 (0.0561 - 0.0760)
Best Fusion: Difficulty-adaptive	0.6029 (0.5710 - 0.6385)	0.0667 (0.0588 - 0.0795)
Blending	0.6028 (0.5689 - 0.6367)	0.0666 (0.0588 - 0.0795)
Edge-enhanced	0.6027 (0.5683 - 0.6385)	0.0666 (0.0582 - 0.0794)
Consistency-weighted	0.6024 (0.5694 - 0.6377)	0.0665 (0.0582 - 0.0792)
Uncertainty-weighted	0.6014 (0.5681 - 0.6368)	0.0663 (0.0582 - 0.0791)
Stacking	0.6008 (0.5670 - 0.6358)	0.0661 (0.0582 - 0.0789)
Confidence-weighted	0.5960 (0.5627 - 0.6322)	0.0657 (0.0577 - 0.0782)
Hierarchical Uncertainty	0.5953 (0.5613 - 0.6312)	0.0658 (0.0578 - 0.0783)
Hierarchical Confidence	0.5953 (0.5613 - 0.6312)	0.0658 (0.0578 - 0.0783)
Neural Network	0.5951 (0.5611 - 0.6314)	0.0655 (0.0574 - 0.0780)

Analysis and Conclusion

- **Overall Trend:** Several dynamic weighting fusion strategies (Difficulty-adaptive, Blending, Edge-enhanced) show a consistent but minor performance improvement over the baseline across both metrics.
- **Best Performance:** The **Difficulty-adaptive** strategy performs best, improving AUROC

by +0.0056 and AUPRC by +0.0010.

- **Statistical Significance:** Despite the improvements, the confidence intervals of all fusion strategies **significantly overlap** with the XGBoost baseline. Therefore, the gains are **not statistically significant**.
- **Recommendation:** Given the extreme difficulty of the task, any stable improvement is worth considering. The **Difficulty-adaptive Fusion** model is the preferred choice, even though its advantage is not statistically certain.

3. Vasopressor Need (vaso) Analysis

Source Data: vaso_confidence_intervals_summary.md

This task represents the most successful application of fusion strategies, clearly demonstrating the value of data complementarity.

Performance Comparison Table: Vasopressor Need (vaso)

Model / Strategy	AUROC (95% CI)	AUPRC (95% CI)
Baseline: XGBoost	0.7500 (0.7206 - 0.7794)	0.2000 (0.1608 - 0.2392)
Baseline: Elastic Net	0.7300 (0.7006 - 0.7594)	0.1800 (0.1408 - 0.2192)
Best Fusion: Neural Network (Basic MLP)	0.7862 (0.7548 - 0.8176)	0.2434 (0.1901 - 0.2967)
Runner-up: Hierarchical (Complexity-based)	0.7861 (0.7549 - 0.8174)	0.2378 (0.1863 - 0.2892)
Ensemble Blending	0.7852 (0.7534 - 0.8170)	0.2447 (0.1920 - 0.2973)
Ensemble Stacking	0.7837 (0.7516 - 0.8158)	0.2462 (0.1931 - 0.2993)
Dynamic Weight (Consistency-weighted)	0.7852 (0.7534 - 0.8170)	0.2446 (0.1920 - 0.2973)
Hierarchical (Three-layer)	0.7853 (0.7535 - 0.8171)	0.2448 (0.1922 - 0.2975)
Neural Network (Deep MLP)	0.7843 (0.7523 - 0.8163)	0.2453 (0.1925 - 0.2982)
Dynamic Weight (Difficulty-adaptive)	0.7811 (0.7491 - 0.8130)	0.2351 (0.1836 - 0.2865)
Dynamic Weight (Edge-enhanced)	0.7712 (0.7374 - 0.8050)	0.2228 (0.1734 - 0.2723)
Hierarchical (Adaptive Threshold)	0.7705 (0.7385 - 0.8025)	0.2126 (0.1642 - 0.2609)
Dynamic Weight	0.7510 (0.7180 - 0.7840)	0.1905 (0.1442 - 0.2368)

(Confidence-weighted)		
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Analysis and Conclusion

- **Overall Trend:** The vast majority of advanced fusion strategies (Neural Networks, Hierarchical, Ensembling) significantly outperform both the XGBoost and Elastic Net baselines.
- **Best Performance:** The **Neural Network (Basic MLP)** and **Hierarchical (Complexity-based)** strategies are top performers, providing a substantial AUROC improvement of $\sim +0.036$ and a corresponding AUPRC increase of $\sim +0.04$.
- **Statistical Significance:** This is a key finding. The lower bound of the best fusion model's AUROC confidence interval (~ 0.755) is higher than the point estimate of the XGBoost baseline (0.750). The limited overlap in CIs strongly indicates the performance gain is both **statistically significant and clinically meaningful**.
- **Recommendation:** **Neural Network or Hierarchical fusion strategies are strongly recommended.** They provide a clear and robust performance advantage for this task.

4. Ventilation Need (vent) Analysis

Source Data: vent_confidence_intervals_summary.md

In contrast to the vaso task, fusion strategies failed to provide a meaningful benefit for predicting ventilation needs.

Performance Comparison Table: Ventilation Need (vent)

Model / Strategy	AUROC (95% CI)	AUPRC (95% CI)
Baseline: XGBoost	0.6939 (0.6598 - 0.7280)	0.1995 (0.1552 - 0.2438)
Baseline: Elastic Net	0.6714 (0.6355 - 0.7073)	0.1507 (0.1215 - 0.1799)
Best AUROC: Hierarchical (Complexity-based)	0.7023 (0.6690 - 0.7356)	0.1967 (0.1517 - 0.2418)
Best AUPRC: Neural Network (Ensemble MLP)	0.6912 (0.6579 - 0.7245)	0.2083 (0.1632 - 0.2534)
Dynamic Weight (Uncertainty-weighted)	0.6989 (0.6656 - 0.7322)	0.2064 (0.1613 - 0.2515)
Ensemble Stacking	0.6982 (0.6649 - 0.7315)	0.1944 (0.1493 - 0.2395)
Neural Network (Deep MLP)	0.6996 (0.6663 - 0.7329)	0.2036 (0.1585 - 0.2486)
Neural Network (Basic MLP)	0.6946 (0.6612 - 0.7279)	0.2080 (0.1629 - 0.2530)
Hierarchical (Adaptive Threshold)	0.6942 (0.6609 - 0.7275)	0.2056 (0.1605 - 0.2507)
Dynamic Weight (Difficulty-adaptive)	0.6938 (0.6604 - 0.7271)	0.2067 (0.1616 - 0.2518)
Hierarchical (Three-layer)	0.6932 (0.6599 - 0.7266)	0.1993 (0.1542 - 0.2443)
Ensemble Blending	0.6904 (0.6570 - 0.7237)	0.2015 (0.1565 - 0.2466)

Analysis and Conclusion

- **Overall Trend:** All fusion strategies perform very closely to the XGBoost baseline, with no strategy showing a clear advantage.
- **Best Performance:** The **Hierarchical (Complexity-based)** strategy is best on AUROC with a negligible +0.0084 improvement. The **Neural Network (Ensemble MLP)** is best on AUPRC with a +0.0088 improvement.
- **Statistical Significance:** The confidence intervals for all fusion models **almost completely overlap** with the XGBoost baseline's CIs. This indicates that the minor observed differences are **not statistically significant**.
- **Recommendation:** The **XGBoost baseline model** is the best choice for this task, as the added complexity of fusion is not justified by the minimal performance change.

5. Length of Stay > 3 Days (los3) Analysis

Source Data: los3_confidence_intervals_summary.md

This is a relatively balanced task where the baseline models perform well.

Performance Comparison Table: Length of Stay > 3 Days (los3)

Model / Strategy	AUROC (95% CI)	AUPRC (95% CI)
Baseline: XGBoost	0.7265 (0.7140 - 0.7397)	0.6627 (0.6461 - 0.6808)
Baseline: Elastic Net	0.7187 (0.7056 - 0.7327)	0.6561 (0.6390 - 0.6738)
Best Fusion: Dynamic Weight (Consistency-weighted)	0.7324 (0.7198 - 0.7452)	0.6714 (0.6543 - 0.6890)
Hierarchical (Complexity-based)	0.7310 (0.7186 - 0.7443)	0.6715 (0.6546 - 0.6889)
Dynamic Weight (Difficulty-adaptive)	0.7311 (0.7187 - 0.7440)	0.6690 (0.6523 - 0.6867)
Hierarchical (Fusion-tendency)	0.7315 (0.7191 - 0.7446)	0.6695 (0.6521 - 0.6871)
Dynamic Weight (Uncertainty-weighted)	0.7310 (0.7186 - 0.7438)	0.6690 (0.6522 - 0.6867)
Hierarchical (Edge-special)	0.7312 (0.7189 - 0.7445)	0.6675 (0.6508 - 0.6846)
Dynamic Weight (Edge-enhanced)	0.7311 (0.7189 - 0.7443)	0.6686 (0.6518 - 0.6863)
Hierarchical (Three-layer)	0.7284 (0.7160 - 0.7416)	0.6668 (0.6499 - 0.6846)
Dynamic Weight (Confidence-weighted)	0.7283 (0.7158 - 0.7414)	0.6663 (0.6497 - 0.6841)
Hierarchical (Adaptive Threshold)	0.7239 (0.7111 - 0.7370)	0.6591 (0.6422 - 0.6765)

Analysis and Conclusion

- **Overall Trend:** Most fusion strategies provide a slight performance improvement over the XGBoost baseline.
- **Best Performance:** **Dynamic Weight (Consistency-weighted)** performs best, with a +0.0059 increase in AUROC and a +0.0087 increase in AUPRC.
- **Statistical Significance:** Despite the consistent improvements, the confidence intervals of all fusion models **highly overlap** with the XGBoost baseline, meaning the gains are **not statistically significant**.
- **Recommendation:** The **XGBoost baseline model** remains a practical and reliable choice.

6. Length of Stay > 7 Days (los7) Analysis

Source Data: los7_confidence_intervals_summary.md

This task is more imbalanced and challenging than los3.

Performance Comparison Table: Length of Stay > 7 Days (los7)

Model / Strategy	AUROC (95% CI)	AUPRC (95% CI)
Baseline: Elastic Net	0.7438 (0.7203 - 0.7682)	0.1859 (0.1650 - 0.2175)
Baseline: XGBoost	0.7413 (0.7178 - 0.7665)	0.1829 (0.1614 - 0.2127)
Best Fusion: Ensemble (Weighted Ensemble)	0.7499 (0.7268 - 0.7747)	0.1945 (0.1717 - 0.2291)
Neural Network (Deep MLP)	0.7498 (0.7267 - 0.7745)	0.1942 (0.1714 - 0.2295)
Hierarchical (Complexity-based)	0.7497 (0.7262 - 0.7748)	0.1924 (0.1704 - 0.2263)
Ensemble Stacking	0.7496 (0.7264 - 0.7742)	0.1942 (0.1714 - 0.2293)
Hierarchical (Fusion-tendency)	0.7496 (0.7261 - 0.7740)	0.1918 (0.1703 - 0.2263)
Ensemble Blending	0.7496 (0.7264 - 0.7748)	0.1940 (0.1710 - 0.2290)
Neural Network (Ensemble MLP)	0.7495 (0.7265 - 0.7743)	0.1937 (0.1711 - 0.2288)
Neural Network (Adaptive MLP)	0.7495 (0.7265 - 0.7743)	0.1937 (0.1711 - 0.2288)
Neural Network (Basic MLP)	0.7494 (0.7263 - 0.7741)	0.1934 (0.1709 - 0.2277)
Neural Network (Wide MLP)	0.7493 (0.7260 - 0.7743)	0.1942 (0.1713 - 0.2288)
Hierarchical (Edge-special)	0.7489 (0.7256 - 0.7739)	0.1891 (0.1676 - 0.2200)

Hierarchical (Three-layer)	0.7472 (0.7226 - 0.7722)	0.1901 (0.1687 - 0.2207)
Ensemble Voting	0.7450 (0.7209 - 0.7698)	0.1843 (0.1641 - 0.2145)
Ensemble (Adaptive Ensemble)	0.7450 (0.7209 - 0.7698)	0.1843 (0.1641 - 0.2145)
Hierarchical (Adaptive Threshold)	0.7431 (0.7193 - 0.7685)	0.1787 (0.1599 - 0.2065)

Analysis and Conclusion

- **Overall Trend:** Similar to los3, nearly all fusion strategies provide a small performance boost over the best baseline model (Elastic Net in this case).
- **Best Performance:** The **Ensemble (Weighted Ensemble)** strategy is the top performer for both metrics, with a +0.0061 increase in AUROC and a +0.0086 increase in AUPRC.
- **Statistical Significance:** Again, the confidence intervals of all fusion strategies **show significant overlap** with the Elastic Net baseline, rendering the improvements **not statistically significant**.
- **Recommendation:** The simpler **Elastic Net baseline model** provides sufficiently effective performance for this task.

Performance Lift Across Tasks

AUROC improvement of the best fusion model compared to the baseline. The effectiveness of fusion varies dramatically across tasks.

